

HW01

ENAE 601: Astrodynamics

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Problem 1**2.7**

Starting with Eq. 2.35a (shown below), prove that $\dot{r} = \mathbf{v} \cdot \hat{\mathbf{u}}_r$ and interpret this result.

$$\mathbf{r} \cdot \dot{\mathbf{r}} = r\dot{r}$$

Beginning with Eq. 2.35a:

$$\mathbf{r} \cdot \dot{\mathbf{r}} = r\dot{r}$$

Note that $r \equiv |\mathbf{r}|$ and $\dot{\mathbf{r}} \equiv \mathbf{v}$:

$$\mathbf{r} \cdot \mathbf{v} = |\mathbf{r}|\dot{r}$$

Rearrange, with $r > 0$:

$$\dot{r} = \frac{\mathbf{r} \cdot \mathbf{v}}{|\mathbf{r}|}$$

Note that $\frac{\mathbf{r}}{|\mathbf{r}|} \equiv \hat{\mathbf{r}}$:

$$\frac{\mathbf{r} \cdot \mathbf{v}}{|\mathbf{r}|} = \hat{\mathbf{r}} \cdot \mathbf{v}$$

From Eq. 2.7:

$$\hat{\mathbf{u}}_r = \frac{\mathbf{r}}{r} = \frac{\mathbf{r}}{|\mathbf{r}|} = \hat{\mathbf{r}}$$

Thus:

$$\dot{r} = \hat{\mathbf{u}}_r \cdot \mathbf{v}$$

The dot product is commutative, so:

$$\dot{r} = \hat{\mathbf{u}}_r \cdot \mathbf{v} = \mathbf{v} \cdot \hat{\mathbf{u}}_r$$

We now have:

$$\dot{r} = \mathbf{v} \cdot \hat{\mathbf{u}}_r \quad \blacksquare$$

The dot product projects the velocity onto the outward radial direction. Thus, $\dot{r}$ is the radial component of velocity: it is positive when the spacecraft moves away from the attracting body, negative when it moves toward it, and zero when the velocity is perpendicular to the radius. A zero radial velocity does not imply that the spacecraft is stationary.

Problem 2**2.8**

Show that $\hat{\mathbf{u}}_r \cdot \frac{d\hat{\mathbf{u}}_r}{dt} = 0$, where $\hat{\mathbf{u}}_r = \frac{\mathbf{r}}{r}$. Use only the fact that $\hat{\mathbf{u}}_r$ is a unit vector. Interpret this result.

Since $\hat{\mathbf{u}}_r$ is a unit vector:

$$\hat{\mathbf{u}}_r \cdot \hat{\mathbf{u}}_r = 1$$

Differentiate both sides with respect to time:

$$\frac{d\hat{\mathbf{u}}_r \cdot \hat{\mathbf{u}}_r}{dt} = \frac{d1}{dt} = 0$$

Apply the product rule:

$$\frac{d\hat{\mathbf{u}}_r}{dt} \cdot \hat{\mathbf{u}}_r + \hat{\mathbf{u}}_r \cdot \frac{d\hat{\mathbf{u}}_r}{dt} = 0$$

The dot product is commutative, so:

$$2\hat{\mathbf{u}}_r \cdot \frac{d\hat{\mathbf{u}}_r}{dt} = 0$$

Thus:

$$\hat{\mathbf{u}}_r \cdot \frac{d\hat{\mathbf{u}}_r}{dt} = 0 \quad \blacksquare$$

A unit vector has constant magnitude, so its time derivative can only describe a change in direction. When nonzero, this derivative is perpendicular to the unit vector. If the direction is constant, the derivative is zero and the result still holds.

Problem 3**2.9**

Show that $v = \left(\frac{\mu}{h}\right)\sqrt{1 + 2e \cos(\theta) + e^2}$ for any orbit.

Beginning with the orbit equation and the specific angular momentum:

$$r = \frac{\frac{h^2}{\mu}}{1 + e \cos(\theta)}, \quad h = r^2 \dot{\theta}$$

Differentiate the orbit equation with respect to θ :

$$\frac{dr}{d\theta} = \left(\frac{h^2}{\mu}\right) \frac{e \sin(\theta)}{(1 + e \cos(\theta))^2}$$

Apply the chain rule and substitute $\dot{\theta} = \frac{h}{r^2}$:

$$\dot{r} = \frac{dr}{d\theta} \dot{\theta} = \left(\frac{h^2}{\mu}\right) \frac{e \sin(\theta)}{(1 + e \cos(\theta))^2} \frac{h}{r^2}$$

Substitute the orbit equation for r and simplify:

$$v_r = \dot{r} = \left(\frac{\mu}{h}\right) e \sin(\theta)$$

The transverse component of velocity is:

$$v_{\perp} = r \dot{\theta} = \frac{h}{r} = \left(\frac{\mu}{h}\right) (1 + e \cos(\theta))$$

Since the radial and transverse directions are perpendicular:

$$v^2 = v_r^2 + v_{\perp}^2 = \left(\frac{\mu}{h}\right)^2 (e^2 \sin^2(\theta) + (1 + e \cos(\theta))^2)$$

Expand and use $\sin^2(\theta) + \cos^2(\theta) = 1$:

$$v^2 = \left(\frac{\mu}{h}\right)^2 (1 + 2e \cos(\theta) + e^2)$$

Take the nonnegative square root:

$$v = \left(\frac{\mu}{h}\right) \sqrt{1 + 2e \cos(\theta) + e^2} \quad \blacksquare$$

This result applies to circular, elliptical, parabolic, and hyperbolic orbits with $h > 0$. Rectilinear motion has $h = 0$, so this expression is undefined for that degenerate case.

Problem 4

2.10

Relative to a nonrotating, earth-centered Cartesian coordinate system, the position and velocity vectors of a spacecraft are $\mathbf{r} = 7000\hat{\mathbf{i}} - 2000\hat{\mathbf{j}} - 4000\hat{\mathbf{k}}$ km and $\mathbf{v} = 3\hat{\mathbf{i}} - 6\hat{\mathbf{j}} + 5\hat{\mathbf{k}}$ km s⁻¹. Calculate:

- The orbit's eccentricity vector
- The true anomaly

Using $\mu = 398\,600 \text{ km}^3 \text{ s}^{-2}$ for Earth.

- The eccentricity vector

Beginning with the magnitudes and the specific angular momentum:

$$r = \sqrt{7000^2 + (-2000)^2 + (-4000)^2} \text{ km} = 8306.624 \text{ km}$$

$$v^2 = (3^2 + (-6)^2 + 5^2) \text{ km}^2 \text{ s}^{-2} = 70 \text{ km}^2 \text{ s}^{-2}$$

$$\mathbf{h} = \mathbf{r} \times \mathbf{v} = (-34000\hat{\mathbf{i}} - 47000\hat{\mathbf{j}} - 36000\hat{\mathbf{k}}) \text{ km}^2 \text{ s}^{-1}$$

The eccentricity vector is:

$$\mathbf{e} = \frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r}$$

Using the vector triple product, this may also be written as:

$$\mathbf{e} = \frac{1}{\mu} \left(\left(v^2 - \frac{\mu}{r} \right) \mathbf{r} - (\mathbf{r} \cdot \mathbf{v}) \mathbf{v} \right)$$

Calculate the scalar product:

$$\mathbf{r} \cdot \mathbf{v} = (7000(3) + (-2000)(-6) + (-4000)(5)) \text{ km}^2 \text{ s}^{-1} = 13000 \text{ km}^2 \text{ s}^{-1}$$

Substitute and simplify:

$$\mathbf{e} \approx 0.288759\hat{\mathbf{i}} + 0.085227\hat{\mathbf{j}} - 0.383986\hat{\mathbf{k}} \quad \blacksquare$$

Its magnitude is $e = |\mathbf{e}| \approx 0.487945$.

- The true anomaly

The true anomaly is measured from the eccentricity vector to the radius vector in the direction of motion:

$$\cos(\theta) = \frac{\mathbf{e} \cdot \mathbf{r}}{er} \approx 0.835592$$

Taking the inverse cosine gives the candidate angle:

$$\theta_0 = \arccos(0.835592) \approx 33.3225^\circ$$

Since $\mathbf{r} \cdot \mathbf{v} > 0$, the spacecraft is moving outward, so $0 < \theta < 180^\circ$. Thus, the correct quadrant gives:

$$\theta \approx 33.3225^\circ \quad \blacksquare$$

Problem 5**2.11**Show that the eccentricity is 1 for rectilinear orbits ($\mathbf{h} = \mathbf{0}$).

Beginning with the eccentricity vector:

$$\mathbf{e} = \frac{\mathbf{v} \times \mathbf{h}}{\mu} - \frac{\mathbf{r}}{r}$$

For rectilinear motion, $\mathbf{h} = \mathbf{0}$:

$$\mathbf{e} = \frac{\mathbf{v} \times \mathbf{0}}{\mu} - \frac{\mathbf{r}}{r} = -\frac{\mathbf{r}}{r} = -\hat{\mathbf{u}}_r$$

Since $\hat{\mathbf{u}}_r$ is a unit vector:

$$e = |\mathbf{e}| = |-\hat{\mathbf{u}}_r| = 1 \quad \blacksquare$$

This holds wherever $r > 0$. Rectilinear motion is a degenerate orbit: although $e = 1$, its specific energy need not be zero, so it need not be an ordinary parabolic trajectory.

Problem 6

2.12

Relative to a nonrotating, earth-centered Cartesian coordinate system, the velocity of a spacecraft is $\mathbf{v} = -4\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - 5\hat{\mathbf{k}} \text{ km s}^{-1}$ and the unit vector in the direction of the radius $\mathbf{r}$ is $\hat{\mathbf{u}}_r = 0.26726\hat{\mathbf{i}} + 0.53452\hat{\mathbf{j}} + 0.80178\hat{\mathbf{k}}$. Calculate:

- The radial component of velocity v_r
- The azimuth component of velocity $v_\perp$
- The flight path angle γ

The supplied unit-vector components are rounded. Their magnitude is approximately 0.999995353, so I normalized them to unit length for the numerical calculation.

- The radial component of velocity

From Problem 1 (2.7):

$$v_r = \mathbf{v} \cdot \hat{\mathbf{u}}_r$$

Substitute the normalized components:

$$v_r \approx \frac{(-4)(0.26726) + 3(0.53452) - 5(0.80178)}{0.999995353} \text{ km s}^{-1}$$

Thus:

$$v_r \approx -3.4744 \text{ km s}^{-1} \quad \blacksquare$$

- The azimuth component of velocity

The speed is:

$$v = \sqrt{(-4)^2 + 3^2 + (-5)^2} \text{ km s}^{-1} = \sqrt{50} \text{ km s}^{-1}$$

Since the radial and azimuth directions are perpendicular:

$$v_\perp = \sqrt{v^2 - v_r^2}$$

Substitute and simplify, retaining full precision in the calculation:

$$v_\perp \approx 6.1586 \text{ km s}^{-1} \quad \blacksquare$$

Here, $v_\perp$ is the nonnegative transverse speed in the orbital plane.

- The flight path angle

Measure γ from the local horizontal, positive toward the outward radial direction:

$$\tan(\gamma) = \frac{v_r}{v_\perp}$$

Thus:

$$\gamma = \arctan\left(\frac{v_r}{v_\perp}\right) \approx -29.4296^\circ \quad \blacksquare$$

The negative radial velocity and flight path angle indicate that the spacecraft is moving toward Earth.

Problem 7**2.13**

If the specific energy ε of the two-body problem is negative, show that m_2 cannot move outside a sphere of radius $\frac{\mu}{|\varepsilon|}$ centered at m_1 .

Beginning with the specific mechanical energy:

$$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r}$$

Since $\varepsilon < 0$, write $\varepsilon = -|\varepsilon|$ and rearrange:

$$\frac{v^2}{2} = \frac{\mu}{r} - |\varepsilon|$$

The squared speed cannot be negative, so:

$$\frac{\mu}{r} - |\varepsilon| \geq 0$$

Therefore:

$$\frac{\mu}{r} \geq |\varepsilon|$$

Since $r > 0$ and $|\varepsilon| > 0$:

$$r \leq \frac{\mu}{|\varepsilon|} \quad \blacksquare$$

At any greater distance, the energy equation would require $v^2 < 0$. Thus, m_2 cannot move outside the stated sphere. Equality requires zero instantaneous velocity; for a nonrectilinear orbit, $h > 0$ prevents this and the inequality is strict.

Problem 8**2.15**

The specific angular momentum of a satellite in circular earth orbit is $60 \cdot 10^3 \text{ km}^2 \text{ s}^{-1}$. Calculate the period.

For a circular orbit, the velocity is perpendicular to the radius and the centripetal acceleration equals the gravitational acceleration:

$$h = rv, \quad \frac{v^2}{r} = \frac{\mu}{r^2}$$

Thus:

$$v = \sqrt{\frac{\mu}{r}}, \quad h = \sqrt{\mu r}$$

Solve for the orbital radius:

$$r = \frac{h^2}{\mu} = \frac{(60 \cdot 10^3 \text{ km}^2 \text{ s}^{-1})^2}{398\,600 \text{ km}^3 \text{ s}^{-2}} \approx 9031.611 \text{ km}$$

The period is the circumference divided by the constant speed:

$$T = \frac{2\pi r}{v} = 2\pi \sqrt{\frac{r^3}{\mu}}$$

Substitute $r = \frac{h^2}{\mu}$:

$$T = \frac{2\pi h^3}{\mu^2}$$

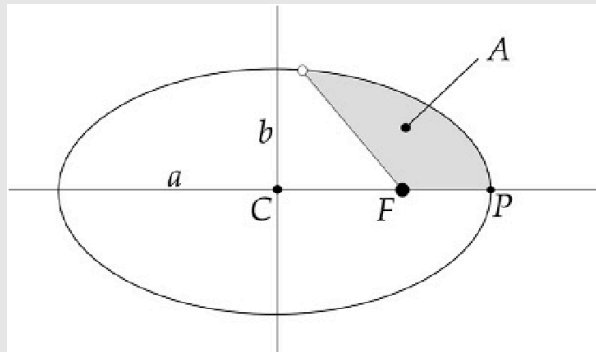
Thus:

$$T \approx 8541.99 \text{ s} \approx 2.372\,77 \text{ h} \quad \blacksquare$$

Problem 9

2.17

Calculate the area A swept out during the time $t = \frac{T}{4}$ since periapsis, where T is the period of the elliptical orbit. See the figure below.



Beginning with the constant areal rate for a two-body orbit:

$$\frac{dA}{dt} = \frac{1}{2} r^2 \dot{\theta} = \frac{h}{2}$$

Measure the swept area from periapsis, so $A(0) = 0$. Integrate over the interval $t = \frac{T}{4}$:

$$A = \int_0^{\frac{T}{4}} \frac{h}{2} dt = \frac{hT}{8}$$

Over one complete period, the radius vector sweeps out the entire ellipse. For semimajor axis a and semiminor axis b :

$$\frac{hT}{2} = \pi ab$$

Thus, one quarter of the period sweeps out one quarter of the total area:

$$A = \frac{\pi ab}{4} = \frac{\pi a^2 \sqrt{1-e^2}}{4} \quad \blacksquare$$

Problem 10**2.22**

The altitude of a satellite in an elliptical orbit around the earth is 2000 km at apogee and 500 km at perigee. Determine:

- The eccentricity of the orbit
- The orbital speeds at perigee and apogee
- The period of the orbit

Using $R_E = 6378 \text{ km}$ and $\mu = 398\,600 \text{ km}^3 \text{ s}^{-2}$ for Earth.

- The eccentricity of the orbit

Convert the given altitudes to distances from Earth's center:

$$r_p = R_E + 500 \text{ km} = 6878 \text{ km}, \quad r_a = R_E + 2000 \text{ km} = 8378 \text{ km}$$

The semimajor axis is:

$$a = \frac{r_p + r_a}{2} = 7628 \text{ km}$$

Since $r_p = a(1 - e)$ and $r_a = a(1 + e)$:

$$e = \frac{r_a - r_p}{r_a + r_p} = \frac{8378 - 6878}{8378 + 6878} \approx 0.098322 \quad \blacksquare$$

- The orbital speeds at perigee and apogee

Apply the vis-viva equation at perigee:

$$v_p = \sqrt{\mu \left(\frac{2}{r_p} - \frac{1}{a} \right)} \approx 7.978\,15 \text{ km s}^{-1} \quad \blacksquare$$

At apogee:

$$v_a = \sqrt{\mu \left(\frac{2}{r_a} - \frac{1}{a} \right)} \approx 6.549\,74 \text{ km s}^{-1} \quad \blacksquare$$

As expected, the spacecraft moves faster at perigee, and $r_p v_p = r_a v_a$ confirms conservation of angular momentum.

- The period of the orbit

For an elliptical orbit:

$$T = 2\pi \sqrt{\frac{a^3}{\mu}}$$

Substitute $a = 7628 \text{ km}$ and the Earth's gravitational parameter:

$$T \approx 6630.21 \text{ s} \approx 110.503 \text{ min} \quad \blacksquare$$